

Atomic layer deposition-based synthesis of photoactive TiO₂ nanoparticle chains by using carbon nanotubes as sacrificial templates†

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Highly ordered and self supported anatase TiO₂ nanoparticle chains were fabricated by calcining conformally TiO₂ coated multi-walled carbon nanotubes (MWCNTs). During annealing, the thin tubular TiO₂ coating that was deposited onto the MWCNTs by atomic layer deposition (ALD) was transformed into chains of TiO₂ nanoparticles (~12 nm diameter) with an ultrahigh surface area (137 cm² per cm² of substrate), while at the same time the carbon from the MWCNTs was removed. Photocatalytic tests on the degradation of acetaldehyde proved that these forests of TiO₂ nanoparticle chains are highly photoactive under UV light because of their well crystallized anatase phase.

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Introduction

Due to its chemical stability, high photoactivity, suitable electronic properties and environmentally friendly nature, the semiconductor TiO₂ has attracted a lot of attention for various applications in modern technologies, *e.g.*, in fuel cells, lithium ion batteries, gas sensors, solar cells, water splitting, and photocatalysis.^{1–6} For most of these applications, nanostructured porous films with large accessible surface areas and optimized phase compositions and morphologies are required.^{7,8} Many methods including electro-chemical anodization, template-assisted electrochemical deposition, sol–gel and hydro/sol–vothermal routes have been employed for the fabrication of TiO₂ nanostructures such as TiO₂ nanospheres, nano-tubes and nanowires.^{9–11}

In this work, we present the controlled synthesis of one-dimensional TiO₂ nanoparticle chains anchored on a Si substrate. Our approach is based on the atomic layer deposition

(ALD) technique. This thin film growth method is known to produce thin, conformal coatings with thickness control down to the atomic level, and has been proven successful for the synthesis and functionalization of nanostructured and highly porous materials.^{12–16} Several authors have reported the ALD-based synthesis of TiO₂ nanotubes by using anodic alumina as a template.^{17–19} Here, we use multi-walled carbon nanotubes (MWCNTs) as sacrificial templates. As illustrated in Scheme 1, a uniform and conformal TiO₂ coating is applied on a dense forest of MWCNTs. Calcination of the ALD-deposited amorphous TiO₂ into the photoactive anatase crystal structure results in burning away the underlying MWCNTs.^{20,21} Simultaneously, a change in morphology of the TiO₂ layer occurs. Upon annealing, the dense TiO₂ coating transforms into interconnected anatase TiO₂ nanoparticles with a morphology of self supporting chains in the same orientation as the original MWCNTs. This morphology is drastically different from the cylindrical hollow tubes as typically obtained after chemical etching away of the anodic alumina templates. Finally, the combination of techniques used in our study results in a thin, porous and completely immobilized layer of TiO₂ nanoparticle chains that are perfectly suited as photocatalyst for applications in air pollution abatement.

Experimental sections

Growth of MWCNTs

The starting substrate was a Si wafer with a diameter of 200 mm. MWCNTs were grown from a 1 nm (nominal) Co catalyst layer. To avoid diffusion of the Co into the Si, a 70 nm TiN was first sputter deposited onto the Si surface (Endura PVD tool, Applied Materials, USA). MWCNTs were grown in a microwave

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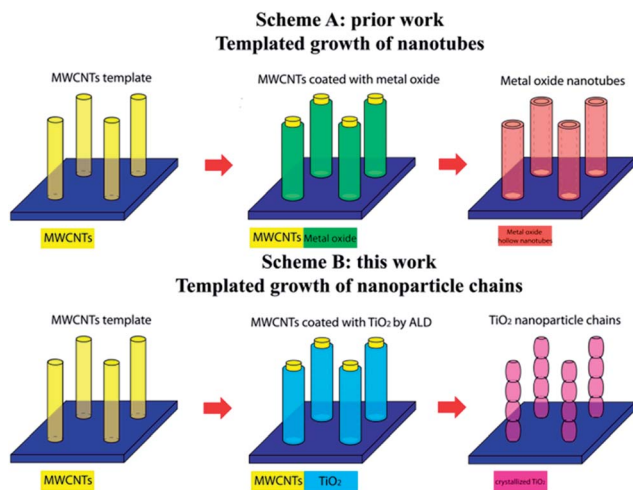
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Scheme 1 Comparative overview of the synthesis route for obtaining metal oxide nanotubes (A) and TiO₂ nanoparticle chains (B) as discussed in this paper. In (A), metal oxide were coated on MWCNTs by various method like sol–gel and electrospray. Metal oxide nanotubes were formed after the removal of the templates. In (B), MWCNTs template is coated with TiO₂ by ALD. Subsequent annealing in air at 500 °C results in the selective removal of the sacrificial CNT template and causes the transformation of the TiO₂ coating into vertical chains of nanoparticles.

(2.45 GHz) plasma enhanced chemical vapor deposition chamber (PECVD, TEL, Japan). In a typical experiment the Co catalyst layer was exposed to a NH₃ plasma for 5 min to transform the film into active metal nanoparticles for CNT growth. Then a C₂H₄–H₂ mixture was flowed into the chamber at a temperature >550 °C for 30 min.²²

TiO₂ ALD on MWCNTs and post deposition annealing

A 5 cm by 1.5 cm piece of grown MWCNTs was loaded into a homemade ALD tool with a base pressure in the low 10^{−7} mbar range, as shown in Fig. S1.†²³ The sample was placed onto a heated chuck, and heated to 100 °C. Tetrakis(dimethylamido) titanium (TDMAT) (99.999% Sigma-Aldrich) and O₃ generated by an ozone generator (Yanco Industries LTD) were alternately pulsed into the ALD chamber at pressures of 0.3 and 0.5 mbar, respectively. The concentration of ozone in the flux was 145 µg mL^{−1}. The pulse (20 seconds) and pump time (40 seconds) were optimized to allow for a uniform coating of TiO₂ along the entire length of the MWCNTs and to prevent the occurrence of chemical vapor deposition type reactions. To remove the carbon, the TiO₂ coated MWCNTs sample was calcined in an oven at 500 °C for 3 hours. The ramp rate was 1 °C per minute and the ambient was air.

Characterizations

In situ and *ex situ* X-ray diffraction characterization (XRD) were measured in a home modified Bruker D8 system (see ESI.†). The morphology and elemental analysis of the cross-sections of the TiO₂ coated MWCNTs were carried out in a scanning electron microscope (FEI Quanta 200 F) equipped with energy dispersive

X-ray analysis (EDX). The bright-field TEM images as well as the high-resolution transmission electron microscopy (HREM) images were obtained using a Tecnai G2 microscope operated at 200 kV. Both a Philips CM 20 and a Tecnai G2 microscope working at 200 kV were employed to get selected-area electron diffraction patterns (EDPs) for analysing the crystallographic phases of TiO₂ nanoparticles. For detailed description of the photocatalysis tests, please see ESI.†

Results and discussions

Uniform ALD on MWCNTs has been reported by depositing metal oxides directly onto the surface of MWCNTs.^{24–27} The defective nature of MWCNTs being synthesized in a plasma environment was reported to enhance the nucleation of ALD processes.²⁸ In addition, it has been demonstrated that the O₃-based process for Al₂O₃ using trimethylaluminum (TMA) yields a better conformality on similar CNT arrays than the TMA–H₂O ALD process.²⁹ Therefore, the choice was made to perform ALD of TiO₂ using TDMAT and O₃ as precursors. Fig. 1(a) and (b) show optical and scanning electron microscopy (SEM) images of a MWCNTs sample after coating with 100 cycles of the TiO₂ ALD process. The transmission electron microscopy (TEM) image in Fig. 1(c) illustrates that the ALD coating resulted in a TiO₂ layer surrounding most of the MWCNTs. A higher resolution TEM image in Fig. 1(d) shows that the MWCNTs were uniformly coated by TiO₂. The original tube width was approximately 10 nm and broadened during ALD to *ca.* 16 nm, meaning that a 3 nm thick TiO₂ layer was deposited onto the MWCNTs, which corresponds to a growth rate of *ca.* 0.03 nm per ALD cycle. An Energy Dispersive X-ray spectroscopy (EDX) characterization was carried out during the SEM measurement to investigate the homogeneity of the TiO₂ coating throughout the entire MWCNTs layer. The inconspicuous variation of the Ti/C EDX

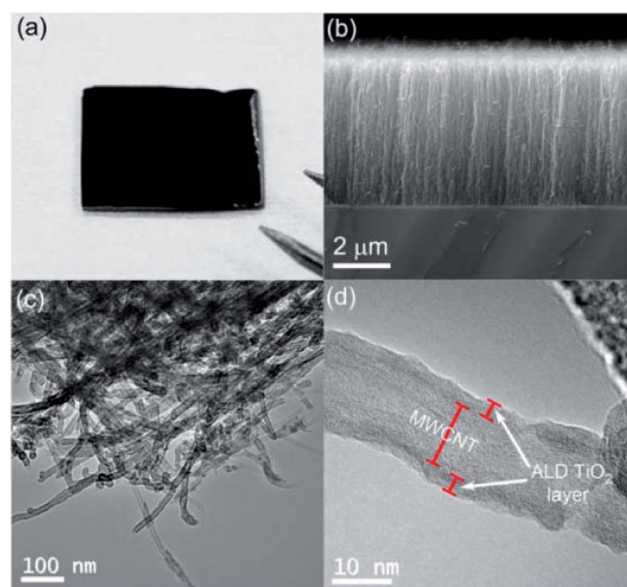


Fig. 1 Optical image (a), SEM image (b) and TEM images (c) and (d) of 6 µm MWCNTs after 100 ALD cycles of TiO₂.

signals ratio measured at distinct intervals along the length of the MWCNTs is indicative of a homogeneous distribution of TiO_2 in the entire layer (see ESI†). A supplementary X-ray fluorescence spectroscopy (XRF) test revealed that the amount of TiO_2 deposited by ALD was about 183 times larger than the amount of TiO_2 deposited on a planar Si substrate that experienced the same ALD process. Provided the growth per cycle on the flat reference sample and on the walls of the MWCNTs was similar, the MWCNTs offered an increase in surface area by a factor of about 183.³⁰ The very thin, uniform, perfectly controllable TiO_2 coating synthesized by ALD on the MWCNTs is strikingly different from other TiO_2 depositions on MWCNTs obtained *via* sol-gel, hydrothermal, solvothermal, electroplating, chemical vapor deposition and other chemical methods, where the coatings are often thicker than 3 nm, not conformal, coarser, uneven or less controllable.^{31–38}

For applications such as photocatalysis and dye-sensitized solar cells, it is essential that the as-deposited amorphous TiO_2 layer is transformed into a crystalline phase.^{39,40} The crystallization of a coated MWCNTs sample upon annealing was studied using *in situ* X-ray diffraction (XRD). The sample was heated in air at a slow ramp rate of 1 °C per minute from room temperature to 850 °C, while irradiating it with a beam of Cu K α radiation and continuously monitoring the diffraction peaks in the range of 15–35° (Fig. 2(a)). From these data it was concluded that crystallization of the TiO_2 layer on the MWCNTs into anatase phase initiates at about 400 °C. Consequently, annealing of the wafer at 500 °C for 3 h should result in a well developed TiO_2 anatase crystal structure that is suited for photocatalysis. The effect of such an annealing step is indicated by the XRD pattern depicted in Fig. 2(b). Note that the observed rutile phase is due to oxidation of the TiN layer beneath the MWCNTs, as further confirmed by a TEM study of the structure.⁴¹ Elemental analysis of the annealed sample indicated that annealing in air resulted in the removal of the carbon from the sample, with only TiO_2 remaining, as illustrated in Fig. 2(c) and (d).

Annealing in air of ultrathin coatings on flimsy supporting frameworks would normally cause a collapse of the structure.⁴² However, due to the conformal nature of the TiO_2 ALD process on the MWCNTs forest, the TiO_2 coating transformed into a self-supporting nanostructure. Fig. 3(a) shows an optical picture of the sample in Fig. 1(a) after annealing in air at 500 °C for 3 hours. Without apparent cracks or wrinkles at the macroscale, annealing clearly resulted in a color change from deep black to shiny purple blue. Given the black color of the MWCNTs, this color change can be related to the removal of carbon, as supported by the EDX results shown in Fig. 2. The cross-sectional SEM picture in Fig. 3(b) is very similar to Fig. 1(b), indicating that the morphology at the microscale was substantially preserved and that calcination did not lead to a collapse of the structure, which is also a valid evidence for the excellent homogeneity of the ALD coating throughout the MWCNTs layer. Further XRF characterization on this sample shows that this annealed porous TiO_2 film has a surface area increase by a factor of 137 compared to the planar substrate (*i.e.* 137 cm² of TiO_2 surface per cm² of Si supporting substrate), showing that

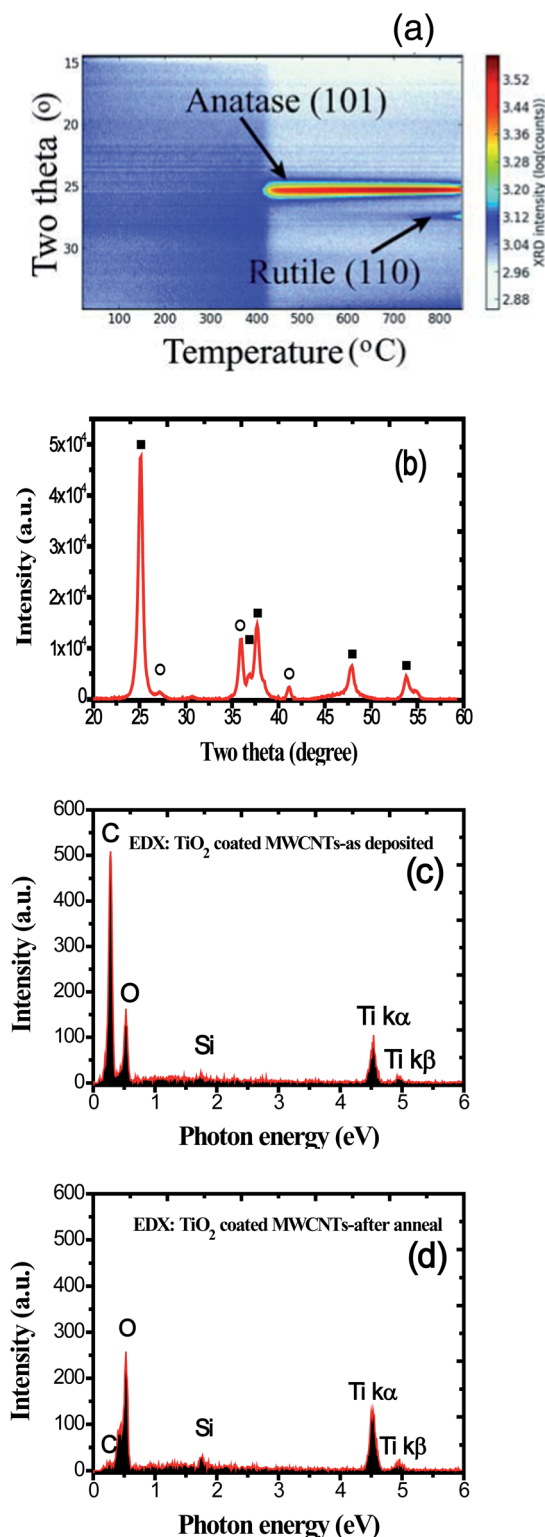


Fig. 2 (a) *In situ* XRD data of TiO_2 coated MWCNTs heated at 1 °C min⁻¹ as a function of the temperature and diffraction angle 2θ . The appearance of anatase and rutile crystal phases is indicated. The colours in the diffraction pattern from blue to red represent the count of diffraction intensity from low to high. (b) XRD spectrum of TiO_2 coated MWCNTs after annealing in air at 500 °C for 3 hours. Anatase (■) and rutile (○) peaks are indicated. (c) and (d) show the EDX characterization of TiO_2 coated MWCNTs as-deposited (c) and after annealing in air at 500 °C for 3 hours (d).

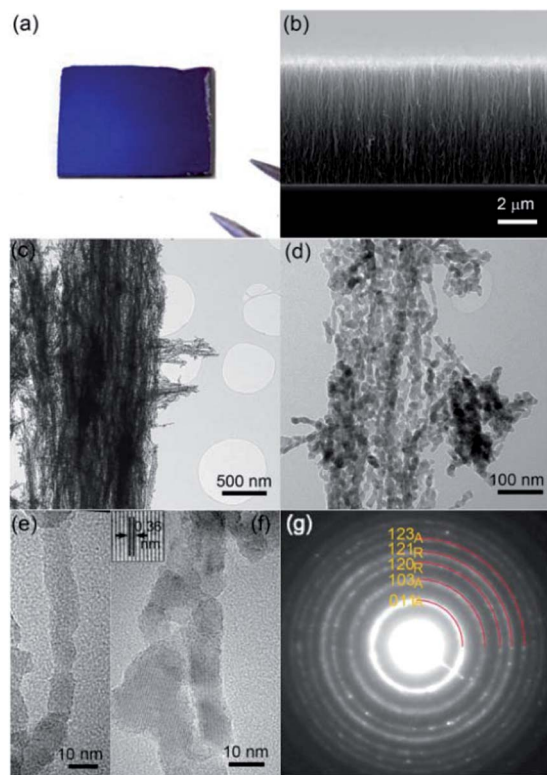


Fig. 3 TiO_2 nanoparticle chains after annealing in air at 500 °C for 3 hours. (a) Optical image, (b) SEM image and (c–f) TEM images of the TiO_2 nanoparticle chains. The removal of the MWCNTs template and the transformation of the TiO_2 coating into nanoparticle chains was clearly revealed from both the low magnification bright-field TEM images in (c) and (d) and the HREM images in (e) and (f). The crystalline phase of the TiO_2 nanoparticles is demonstrated by the HREM images in (e) and (f) and the selected-area electron diffraction pattern in (g). The diffraction rings in (g) could be indexed both using the anatase and rutile structure, with a subscript A and R referring to anatase and rutile, respectively.

this porous film not only inherits the forest-like morphology of the MWCNTs template at the microscale, but also retains as much as 75% of the surface area of the original MWCNTs forest.

A more detailed investigation of the structure was performed using TEM. This study showed that calcination introduced a remarkable change in the morphology at the nano scale. During annealing the TiO_2 coating that originally surrounded the MWCNTs gradually transformed into a network of interconnected chains of nanoparticles (Fig. 3(c) and (d)). A high resolution electron microscopy (HREM) image of such a particle revealed its crystalline nature (Fig. 3(e)). A lattice spacing of 0.36 nm was distinguished, corresponding to the distance between anatase (011) planes (Fig. 3(f)). The selected area electron diffraction pattern in Fig. 3(g) indicates diffraction of (011)A, (103)A, (120)R, (121)R and (123)A crystal planes originating from both the anatase and rutile phase. A HREM study indicated that the TiO_2 nanoparticles only appear in the anatase phase, whereas the rutile phase could be related to oxidized TiN that was scratched from the surface during TEM sample preparation. The particles are fully crystalline throughout their entire cross-section and the mean TiO_2 particle size is *ca.* 12 nm

(see ESI†). We suggest that the chain-like structure in the vertical direction could be formed because it originates from an ultrathin but continuous and conformal TiO_2 layer. When the continuity of the layer would be interrupted at several locations along the tubes, a substantial collapse of the structure is expected upon removal of the template. Furthermore, TiO_2 deposition at the numerous touching zones of the closely packed, flexible MWCNTs on the wafer is responsible for the lateral strengthening of the structure. It is worth noting that the formation of these metal oxide nanoparticle chains would likely not happen when either the metal oxide layer is too thick (>30 nm) or the diameter of the MWCNTs is too large.^{24,43,44} Besides, by using ALD some groups reported that hollow metal oxides nanotubes or nanoribbons were formed when the organic templates like nanocellulose aerogels and peptides were removed by calcinations.^{45,46} We postulate that during annealing in air, heat transmitted by the MWCNTs locally sintered the TiO_2 thin film and transformed the tubular coating into chains of nanoparticles after the MWCNTs were removed. In order to investigate the effect of annealing ambient, a control sample after the same ALD process was annealed in He ambient at 600 °C for 3 hours. After annealing, as shown by the TEM picture in Fig. 4(a), the TiO_2 thin coating on MWCNTs transformed into particles surrounding the MWCNTs, which remained intact during the annealing in inert ambient. However, due to the lack

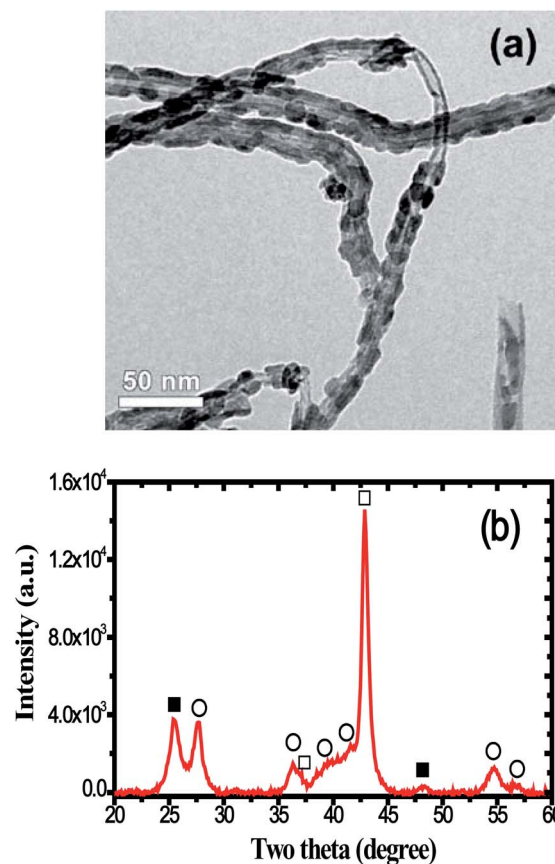


Fig. 4 (a) TEM picture and (b) XRD spectrum of the control sample annealed in He ambient at 600 °C for 3 hours. Anatase (■), rutile (○) and TiN (□) peaks are indicated.

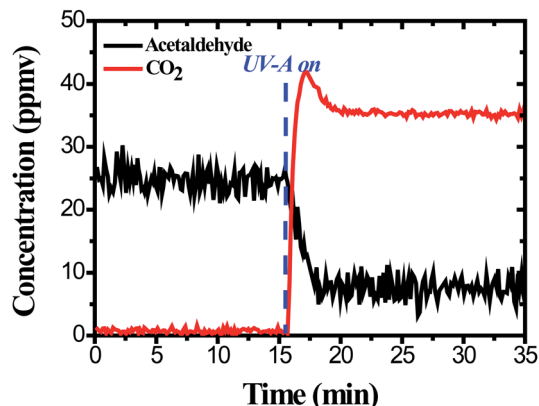


Fig. 5 Reactor outlet concentrations of acetaldehyde (black curve) and CO_2 (red curve) during the photocatalytic test on TiO_2 nanoparticle chains. The moment the UV-A lamp is switched on is indicated with the dotted blue line.

of oxygen and the carbonaceous template, only few TiO_2 nanoparticles were crystallized into the anatase phase while other TiO_2 nanoparticles preserved their amorphous nature.^{47,48} In Fig. 4(b), XRD characterization on this sample further proved that the anatase [101] peak at 25.2° is much weaker compared to the sample annealed in air shown in Fig. 2(b). The rutile peaks of TiO_2 apparent in Fig. 4(b) are related to the annealing of the TiN layer that is present underneath the MWCNTs in He ambient at high temperature.⁴⁹

The obtained porous film, consisting of anchored anatase nano-particle chains, exhibits several merits for gas phase photocatalysis. The rigid structure provides a large surface area and a sufficiently porous arrangement to allow gas molecules to diffuse, adsorb and react on the active surfaces. Moreover the direct synthesis route of fixed TiO_2 nanoparticle chains offers an important benefit over common nanosized anatase powder particles that need additional immobilization efforts. Due to their nano size, nonfixed powder particles have the potential risk of jeopardizing human health.⁵⁰ As a control, a sonication treatment was applied to our film which proved that it is mechanically robust (see ESI†). These films could for instance have practical applications in the field of photocatalysis. As a proof of concept, a feasibility experiment involved testing a wafer of dimensions 5 cm by 1.5 cm towards photocatalytic acetaldehyde degradation in the gas phase. Acetaldehyde is an important source of indoor air pollution. Air spiked with 25, 52 and 80 ppmv of acetaldehyde was flowed over the film at a total flow rate of $400 \text{ cm}^3 \text{ min}^{-1}$. After the adsorption-desorption equilibrium was established, the UV-A lamp was switched on. The evolution of the acetaldehyde concentration, as well as that of CO_2 as the degradation product was continuously monitored using on-line FTIR spectroscopy. More details on the photocatalytic test can be found in the ESI.†

The change in the reactor outlet concentrations of acetaldehyde and CO_2 for an inlet concentration of 25 ppmv is displayed in Fig. 5. From this it can be concluded that the moment the UV-A lamp is switched on, acetaldehyde is degraded into CO_2 (and water, not shown). The acetaldehyde degradation was

determined to be 72%, 45% and 30% for the 25, 52 and 80 ppmv inlet concentrations, respectively. CO_2 is the main degradation product, which is evidenced by a carbon balance close to 100%. However, contrary to the TiO_2 nanoparticle chains with anatase phase, the control sample annealed in He showed no photo-activity at all, which is due to the low proportion of crystalline nanoparticles in the film, as demonstrated by the TEM and XRD results discussed before.

Conclusions

In conclusion, we introduced a novel ALD-based method to synthesize a forest of 1D anatase TiO_2 nanoparticle chains by using MWCNTs as sacrificial templates. The resulting self supported structure of fully crystallized nanoparticles offers a porous network with a large surface area, which is a desirable property for applications in e.g. photocatalysis, sensing and photovoltaics. Photocatalytic tests showed efficient degradation of acetaldehyde, a ubiquitous pollutant in indoor air, thus demonstrating that the self supported TiO_2 nanoparticle chains can be readily employed as an effective photocatalyst.

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